

Topology-Correcting Differentiable Triangle-Soup Reconstruction via Persistent Homology

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Abstract

Resampling priors can steer where a differentiable triangle-soup reconstruction spends its triangle budget, but our controlled study of that channel (supplementary material) found their topological value carried largely by spatial width, and no prior shape repairs one-dimensional features: concentrated priors manufacture phantom handles, spread priors never beat a random control. We move topology from the allocation channel into the objective: $L = \text{photometric} + \lambda(t) L_{\text{topo}}$, where L_{topo} is an optimally-matched distance between the live surface’s persistence diagram and a fixed target diagram. Gradients flow through a pair-frozen backward pass: matched birth/death simplices are re-expressed as closed-form circumradii of live sample positions — exact for Gabriel simplices, which constitute all significant pairs in practice — and a recruitment term restores the gradient optimal matching provably lacks when a target feature is missing. A single ratio knob ρ sets the loss scale by gradient-norm calibration; a curriculum proved unnecessary. On budget-starved synthetic scenes the loss is topology-specific where a norm-matched non-topological control is not: enclosed voids improve 4.0–7.9× and — the channel’s distinguishing result — torus loops improve 2.3× with zero phantom handles across seeds, while the control collapses a loop. The channels compose: loss plus the best prior reaches 7.6× below baseline at the best Chamfer. Component counting (H0) is again matched by the generic control — this class needs no topological information. All verdicts replicate, with no per-shape tuning, on four external genus-known meshes — one a raw artist mesh ingested and certified rather than constructed — at 2.1–10.4× and equal-or-better Chamfer, and the loss degrades gracefully (never below baseline) under sensor noise up to one sample spacing. Prescription: correct topology in the loss, allocate with a wide prior, and combine.

1. Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [28], periodically resampling triangles under a fixed budget. Photometric losses are topologically blind: reconstructions that differ in genus, connectivity, or enclosed-void structure can be indistinguishable to Chamfer and Hausdorff distances while the bottleneck distance between persistence diagrams separates them by an order of magnitude (suppl. Table S1). We first attacked this through the allocation channel — biasing where the resampler spends triangles — in a controlled study reported in the supplementary material (§A). The verdict was mixed: a topological prior helps enclosed voids, but most of the gain is reproduced by any equally-wide non-topological prior, and no prior shape repairs one-dimensional features: concentrating manufactures phantom handles, and spreading never beats a random control on loops.

This paper moves topology into the objective: a differentiable persistence term in the training loss, so gradients act on the birth/death values of the reconstruction’s topological features directly, rather than on where triangles respawn. Concretely:

- A matched persistence loss with a pair-frozen backward (§3): the forward pass is ordinary exact persistence (GUDHI alpha complex); the backward pass re-expresses each paired birth/death simplex as the closed-form circumradius of live surface samples; 100% of significant pairs are Gabriel-exact on our shapes, so the frozen gradient is the true gradient almost everywhere.
- A recruitment term for missing features: optimal partial matching yields exactly zero gradient when a target feature has no live counterpart — the most important failure mode. Recruitment attaches the nearest live bar instead, restoring a descent direction; a controlled toy probes its known location-blindness, and an in-loop ablation shows the term is

load-bearing — removing it collapses the loop-class win to baseline (§5).

- One calibrated knob: λ is set by a gradient-norm ratio ρ against the photometric loss; the response is flat over a $10\times$ range, and a curriculum adds nothing once calibrated — an ablation, not an assumption.
- Channel-controlled evaluation: every claim is tested against a norm-matched non-topological control acting through the identical channel (a repulsion on the same surface samples), inheriting the allocation study’s control discipline.

The loss is topology-specific for enclosed voids ($4.0\text{--}7.9\times$ error reduction at Chamfer parity) and, unlike every prior shape we tested, for loops ($2.3\times$, zero phantom handles across seeds). The two channels compose (best-of-both $7.6\times$); component structure (H0) is again matched by the generic control — three channels agree it carries no exploitable topological signal.

2. Related work

Differentiable rasterization and soup pipelines. OpenDR [22], neural mesh rendering [17], SoftRas [21] and nvdiffrast [19] established gradient flow from pixels to geometry; Gaussian splatting [18] and triangle splatting [13] trade connectivity for speed. We build on DiffSoup [28], whose in-loop prune/respawn resampler is exactly the mechanism our allocation-channel study (suppl. §A) steered with importance priors; here the resampler stays untouched and the objective changes. Vertex-domain optimizers such as large-steps preconditioning [24] are complementary — they change how vertices move, we change why.

The allocation channel: why priors alone fail. Three findings of our allocation-channel study set this paper’s problem. (1) Geometric metrics are topology-blind: on bisection-matched probes with Chamfer equal by construction, the discriminating-dimension bottleneck separates correct from broken topology by $35\text{--}40\times$ (suppl. Table S1). (2) For voids, a spread topological prior cuts the bottleneck $33\text{--}53\%$ below baseline — but a width-matched random control recovers most of that gain, leaving only a small, shape-dependent topological residual: the prior’s value is carried largely by its width. (3) No prior shape repairs loops: the best torus arm is the non-topological wide control ($.0267$ vs the topological spread prior’s $.0316$), and the concentrated topological field manufactures phantom handles (4.4 significant loops against the true 2). The composition condition C5 reuses that study’s strongest field (B4; suppl. §A) — though any width-matched allocation prior could fill the slot.

Persistence in learning objectives. Persistent homology [11] with its stability guarantees [8] has been used as a trainable signal via differentiable sorting of filtration values [5], for topology-preserving image segmentation [7, 15], point-cloud and shape optimization [4, 26], and diagram-metric optimization with convergence analysis [6]; barcode-valued maps have since been given a general differential framework [20], and the per-step cost of persistence-guided descent attacked directly [25]. Our backward pass is in this family — unlike work that back-propagates through filtration-value sorting or barcode coordinates in general position, we re-express each paired simplex’s birth/death as a closed-form circumradius of live surface samples — with three pipeline-specific ingredients. (i) Alpha-complex pairing over $\alpha\times$ area surface samples of a live triangle soup, rather than image grids or raw clouds, couples the diagram to the geometry the renderer is actually training. (ii) A Gabriel-exactness gate: the circumradius re-expression equals the filtration value only for Gabriel simplices, so we verify it per pairing (relative 10^{-6}) — orthogonal to the standard pair-frozen device: it certifies the frozen gradient rather than assuming it. (iii) Recruitment for missing features: optimal partial matching has zero gradient exactly when a target feature is entirely absent — the worst failure mode — so we deliberately trade metric fidelity for a restored descent direction (§3), and quantify both its known pathology and its necessity (§5).

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [2, 16, 27], and budgeted remeshing has a long lineage [1, 3, 9, 12, 14]; our contribution is orthogonal — the budget mechanics stay DiffSoup’s, only the training signal becomes topology-aware.

3. Method: a matched persistence loss for triangle soups

3.1. Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary + $0.8 L_1 + 0.2 \text{SSIM}$) [28], which we adopt unchanged: every ablation below isolates the topology channel, never photometric tuning. We add

$$L = L_{\text{photo}} + \lambda(t) L_{\text{topo}}(V), \quad (1)$$

where V are the soup’s vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed target bundle $\mathcal{T}$: the significant finite bars of the ground-truth shape’s diagram, computed once at the same sampling density M the loss will see (alpha values shift with point spacing: density matching is

176 a correctness requirement) and normalized by the tar-
 177 get’s bounding-box diagonal, which is locked for all live
 178 evaluations.

179 3.2. Live diagram and pair-frozen backward

180 At each refresh we draw M surface samples ($M=2048$
 181 unless the measurement-floor rule of §4 dictates a
 182 denser bundle) $X_j = \sum_i w_{ij} V[F(f_j), i]$ with faces
 183 f_j drawn $\sigma(\alpha) \times \text{area-weighted}$ and barycentric weights
 184 w_{ij} frozen; X is recomputed from live V every step, so
 185 gradients reach vertices while the combinatorics stay
 186 fixed. The alpha complex of X [10], computed exactly
 187 with GUDHI [23], yields persistence pairs (birth sim-
 188 plex, death simplex) per dimension. For a Gabriel sim-
 189 plex the alpha filtration value is its squared circumra-
 190 dius, so each paired bar’s coordinates are closed-form
 191 functions of a few sample positions, all exactly differ-
 192 entiable:

$$193 \quad R_{\text{edge}} = \frac{1}{2} \|p_1 - p_0\|, \quad R_{\text{tri}} = \frac{\|u\| \|v\| \|u-v\|}{2 \|u \times v\|}, \quad (2)$$

$$R_{\text{tet}} = \|y\| \quad \text{with} \quad Ay = \frac{1}{2} \text{diag}(AA^T),$$

194 where $u = p_1 - p_0$, $v = p_2 - p_0$, and A stacks the
 195 rows $p_i - p_0$ (the circumcenter is $p_0 + y$). Edges carry
 196 H0 deaths and H1 births, triangles H1 deaths and H2
 197 births, tetrahedra H2 deaths. We verify, per pairing,
 198 that the re-expressed value matches the filtration value
 199 (relative 10^{-6}); a pair failing this Gabriel test is de-
 200 tached — it keeps its (correct) value in the loss but
 201 propagates no gradient through that simplex, so an ex-
 202 actness failure can never inject a wrong gradient; the
 203 worst case is a missing tug for one refresh interval. In
 204 practice the gate never fires: 100% of significant pairs
 205 are Gabriel-exact on clean clouds across all six bench-
 206 mark shapes, with zero failures in the stress test’s 2,400
 207 noisy refreshes (§5). The per-refresh failure count is
 208 logged, and stronger fallbacks (dropping failing pairs,
 209 subgradient surrogates, locally raising M) remain com-
 210 patible with the gate. The pairing is treated as locally
 211 constant — the standard pair-frozen device [5, 6] — re-
 212 freshed every $K=10$ steps and, necessarily, after every
 213 resampling event, which replaces the vertex and face
 214 tensors wholesale.

215 3.3. Matching, diagonal pressure, and recruitment

216 Let D be the live significant bars and $\mathcal{T}$ the target
 217 bars of one dimension. An optimal partial matching
 218 (Hungarian on the standard augmented cost; equiva-
 219 lently W_2) splits bars into matched pairs, unmatched

live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j) \text{ matched}} [(b_i - b_j^*)^2 + (d_i - d_j^*)^2] \quad (3)$$

$$+ \sum_{i \text{ unmatched}} \left(\frac{d_i - b_i}{2} \right)^2 + L_{\text{recruit}}.$$

222 The first term pulls matched features to their target
 223 coordinates; the second pushes spurious features to the
 224 diagonal. The third exists because optimal matching
 225 has a blind spot: when a target feature is entirely miss-
 226 ing, the cheapest transport diagonalizes both sides and
 227 the gradient is exactly zero. Recruitment restores a
 228 descent direction. Let $\mathcal{T}_{\text{miss}}$ be the unmatched target
 229 bars, processed in decreasing persistence order, and let
 230 the candidate pool $\mathcal{A}$ hold every live bar not claimed
 231 by the matching — unmatched significant bars and,
 232 crucially, sub-threshold bars, which matching never
 233 sees. Each missing target greedily claims its nearest
 234 pool bar, without replacement, under the same squared
 235 birth-death metric as the matched term:

$$L_{\text{recruit}} = \sum_{j \in \mathcal{T}_{\text{miss}}} [(b_{k(j)} - b_j^*)^2 + (d_{k(j)} - d_j^*)^2], \quad (4)$$

$$k(j) = \arg \min_{k \in \mathcal{A}_j} [(b_k - b_j^*)^2 + (d_k - d_j^*)^2],$$

237 where $\mathcal{A}_j$ is the pool remaining after earlier recruit-
 238 ments; recruited bars are excluded from the diago-
 239 nal term. Both coordinates of the recruited bar back-
 240 propagate through the same circumradius machinery as
 241 matched bars. This deliberately departs from a metric-
 242 faithful W_2 ; §5 probes its known pathology (location-
 243 blindness) directly, and the C6 ablation shows the term
 244 is load-bearing in training.

245 3.4. Calibration instead of curriculum

246 $\lambda(t)$ ramps linearly from 20% to 50% of training, with
 247 peak set once by a gradient-norm ratio at the first ac-
 248 tive step: $\lambda_{\text{peak}} = \rho \|\nabla_V L_{\text{photo}}\| / \|\nabla_V L_{\text{topo}}\|$. The sin-
 249 gle knob ρ is dimensionless; $\rho=0.1$ throughout. The
 250 tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, and an
 251 uncalibrated-schedule ablation (constant λ from step
 252 one) matches the ramped version — calibration, not
 253 scheduling, is what makes the knob safe. The window
 254 itself carries no measurable sensitivity either: alterna-
 255 tive ramps (10–40% and 30–60% of training; sphere,
 256 three seeds) land at $.0154 \pm .0017$ and $.0179 \pm .0019$
 257 against the default window’s $.0156 \pm .0013$ — within
 258 seed noise; the 20:50 default is a conservative choice,
 259 not a tuned hyperparameter.

260 3.5. Conditions

261 C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2:
 262 $+a$ short-range repulsion on the same frozen samples,

calibrated with the same ρ — a norm-matched non-topological control through the identical channel (C2g: a gentler radius). C3: C1 without the ramp. C5: C1 + the best prior of our allocation study (the spread topological field B4; suppl. §A). C6: C1 with the recruitment term removed (pure matching + diagonal, all else identical) — does recruitment matter in-loop, or only for cold-start repair? The hook adds one loss line to the training loop; flag off, no new code executes, and a $\rho=0$ run is divergence-equivalent to baseline under CUDA nondeterminism.

4. Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (every-8th test holdout $\Rightarrow$ 21–63 training views): sphere and cube (void class H2, $\beta=(1,0,1)$, budget $N=1200$), torus (loop class H1, $(1,2,1)$, $N=700$), two spheres (component class H0, $(2,0,2)$, $N=700$), and genus-2 double torus as a stretch case $((1,4,1)$, $N=2000$). Budgets are the allocation study’s headroom points (suppl. §A): tight enough that baseline topology is imperfect, so there is something to win. Four external, genus-known meshes probe generality beyond the analytic family (§5.2). Three are watertight benchmarks: spot (genus 0) and bob (genus 1) from the Crane model repository (CC0), and the fandisk CAD benchmark (genus 0) — rendered with the same pipeline (48 views, 200 px) at unchanged class budgets; a seed-0 C0 pilot confirms baseline imperfection (tail bottleneck .043–.054, inside the analytic band). The fourth, the tom-yum pot (Fig. 2), probes the step every real deployment needs: a raw artist-authored mesh rather than a clean benchmark. As exported from Blender it is textbook triangle soup — 6,318 vertices in 9 overlapping open shells, mixed quad/12-gon faces, 50 boundary edges after exact welding, 232 non-manifold junction edges — with no trustworthy ground-truth topology (the same defect that excluded ShapeNet). An ingest-and-certify pass makes it usable: exact-duplicate weld, offset-solidification into a thin closed shell (exact unsigned distance field + marching cubes at iso ε), dropping 804 enclosed pocket shells, then certification by measurement — closed, edge-manifold, consistently-oriented edge certificate, exact simplicial homology, a watertightness cross-check, and bit-identical rebuilds — yielding one body of genus 3, $\beta=(1,6,1)$ exact (190,990 vertices), rendered and budgeted as a void-class shape ($N=1200$, 48 views, 200 px). Where the benchmark trio’s topology is known by construction, the pot’s is known by certificate — the pipeline real scans will need. The pot also exercises the measure-

ment floor of §6 prospectively: no H1 loop of the certified shell (handle bores, flue) clears the $6r_{\text{med}}$ floor at any working M , while the narrow flue mouth caps at small α , making the flue interior an enclosed chamber — an H2 void on the pot axis. The void first clears the floor at $M=8192$ (margin $1.20\text{--}1.23\times$ across five sampling seeds), so the pre-registered floor rule builds its target bundle there — the study’s only protocol deviation — and the loss samples the live surface at the same density (the density-matched contract). The bundle carries exactly one significant bar (H2), stable to re-sampling ($\sim 10^{-5}$).

Protocol. 2,500 steps; resampling every 100 steps (untouched); $\rho=0.1$, ramp 0.2:0.5, refresh $K=10$, $M=2048$; decisive shapes (sphere, torus) run 5 seeds with the full condition set (C0/C1/C2/C3/C5, plus the recruitment ablation C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2), double torus 2 seeds. A sensor-noise stress test (C7/C7h = C1 + noise; decisive shapes, 3 seeds) perturbs the cloud the loss plan sees with i.i.d. Gaussian noise at every refresh, $\sigma = 0.5\%/1\%$ of the target diagonal ($\approx 0.5/1.0\times$ the median sample spacing at $M=2048$): pairing, Gabriel gates, and recruitment decide on the noisy observation; the pulls apply to the true live positions. The torus restricts the loss to H1: its own void bar is sub-threshold at $M=2048$, and a loss must not act on features it cannot reliably measure at its sampling density. For the same reason the double torus targets its two measurable loops (of four) — matching what the reconstruction itself can express at any feasible budget. The generality wave runs C0/C1/C2 with three seeds; bob, unlike the thin torus, has both its loop pair and its void significant at $M=2048$ (fat tube), so it runs the full unrestricted bundle; the tom-yum pot runs at the floor-rule density above.

Metrics and verdict. Primary: bottleneck distance to the target diagram in the shape’s discriminating dimension, tail-averaged over the last trajectory dumps, seed-averaged (20k evaluation samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. Verdict rule: C1 must beat C0 and every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not a topological win. Chamfer parity, formally: the passing arm’s seed-mean Chamfer must satisfy $\text{Ch}(C1) \leq 1.15 \text{Ch}(C0)$; the 15% headroom absorbs seed variance and CUDA nondeterminism and in practice never did any work — every passing arm’s Chamfer was at or below baseline. CUDA training is not bit-reproducible (baseline re-runs diverge at the

first resample), so no bit-level claims are made anywhere.

5. Results

Before integration we verified a gate: the loss alone (no photometric term, Adam on point positions) repairs defective clouds on all four controlled probes — a punctured sphere’s void closes ($8.8\times$), a spurious handle dies at the diagonal, mis-spaced components reach target separation ($1229\times$), and — the recruitment probe — a bridged merge with no significant live H0 bar is severed ($450\times$, $\beta_0 1\rightarrow 2$), the case where optimal matching provably has zero gradient. The predicted location-blind pathology did not materialize (mechanism and figure: suppl. §C, Fig. S4).

5.1. The C-matrix

Table 1 reports tail bottleneck (mean $\pm$ sd over seeds) in each shape’s discriminating dimension; Chamfer in the text where it matters; Fig. 1 shows the full training trajectories for the two decisive shapes.

Voids (H2): topology-specific, large. Sphere $4.0\times$ (16.2σ Welch vs C0), cube $7.9\times$ (18.5σ), both at equal-or-better Chamfer (sphere 0.46 vs 0.50 ; cube 0.92 vs 1.05). The norm-matched control is not merely weaker — it is destructive: it pins the void’s death at its repulsion equilibrium ($2.2\times$ worse than baseline, identical across seeds) and on the cube erases the void outright. Nor is this control-strength dependent: a gentle variant at half the push radius still lands $1.6\times$ worse ($.0972$ vs $.0623$) — the ordering is $C1 < C0 < C2g < C2$, so no tested strength reproduces any of the gain (per-seed spread: suppl. Fig. S3; pin-and-drag mechanism: §6).

Loops (H1): the channel’s distinguishing result. In the allocation-channel study (suppl. §A) no prior shape won this class — concentration manufactured phantom handles; spreading never beat a random control. The loss cuts torus H1 error $2.3\times$ with zero phantom handles in all five seeds (significant-count trajectories flat at the true $\beta_1=2$, suppl. Fig. S5), while the control collapses one loop by step ~ 800 and never recovers it. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not: a loop has minimal spatial support, so allocation can only over- or under-tessellate its neighbourhood (manufacturing phantoms or thinning the tube), while value pressure moves, merges, or tightens the loop directly (§6). Voids, whose support is the whole shell, respond to both channels — why the composition (C5) wins everywhere.

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error $9.5\times$, but the generic control matches it on the diagram metric ($.0008$ vs $.0007$); the topological increment appears only in Chamfer (0.54 vs 1.94). With the allocation-channel results, this is a statement about the feature class: β_0 structure is carried by the count and spatial separation of shells — properties any spreading pressure improves and the photometric term already infers from images. Topological guidance earns its keep where geometry and topology decouple: loops and voids.

Composition (C5). Loss + spread prior is best everywhere tested — sphere $.0082$ ($7.6\times$ vs C0, $\approx 2\times$ vs C1 alone, $\approx 4.4\times$ below the prior alone), torus $.0133$ — with the best sphere Chamfer as well; on the torus its Chamfer ($.502$) trails only the no-ramp arm ($.495$). Allocation and value-tuning are complementary mechanisms, not substitutes.

Stretch case (genus 2): saturated, honestly. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly $.0264$ (sd 0 over seeds), and Wasserstein is equally pinned ($.0527$ – $.0528$): the two tube loops are unexpressable at feasible budgets (replicating the allocation study’s floor, suppl. §A), and they dominate both diagram metrics regardless of guidance. The loss adds no phantoms there (β_1 -significant count = 2 in every run) and its Chamfer is marginally best; the case measures the representation’s ceiling, not the channel’s — both floors are quantified in §6.

Ablations. ρ -response flat over $\{0.03, 0.1, 0.3\}$ ($.0181/.0172/.0177$ in the matched seed-0 sweep on the sphere; the five-seed $\rho=0.1$ arm is Table 1’s C1); C3 (no curriculum) $\approx$ C1 on both decisive shapes and off the analytic family — on the genus-1 external mesh C3 lands at $.0171\pm.0026$ vs C1’s $.0208\pm.0008$, so the curriculum-free finding generalizes. Overhead, honestly: the loss adds a fixed ≈ 46 s per 2,500-step run (refresh 174ms at $M=2048$ every 10 steps) — more than the 33–41s photometric baseline on these deliberately tiny scenes, but the cost scales with M ($O(M \log M)$ per refresh), not scene or image complexity, so it amortizes where rendering dominates; no trainable parameters are added.

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal, all else identical) collapses the torus win entirely: $.0411\pm.0038$, statistically baseline (0.6σ from C0, 12.4σ worse than C1) at baseline Chamfer — though the

Table 1. Tail bottleneck to target (lower is better); $\times$ = reduction vs C0. Verdict requires $C1 < C0$ and $C1 <$ every control at Chamfer parity.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior
sphere (H2)	.0623 $\pm$.0063	.0156 $\pm$.0013 (4.0 $\times$)	.1372 (worse)	.0151 $\pm$.0004	.0082 $\pm$.0007 (7.6 $\times$)
cube (H2)	.0582 $\pm$.0046	.0074 $\pm$.0011 (7.9 $\times$)	.1346 (β_2 1 $\rightarrow$ 0)	—	—
torus (H1)	.0424 $\pm$.0031	.0180 $\pm$.0016 (2.3 $\times$)	.0457 (β_1 2 $\rightarrow$ 1)	.0155 $\pm$.0015	.0133 $\pm$.0004 (3.2 $\times$)
two spheres (H0)	.0065 $\pm$.0008	.0007 $\pm$.0002 (9.5 $\times$)	.0008 (ties C1)	—	—
double torus (H1)	.0264 $\pm$.0000	.0264 $\pm$.0000 (saturated)	.0264 $\pm$.0000	—	—

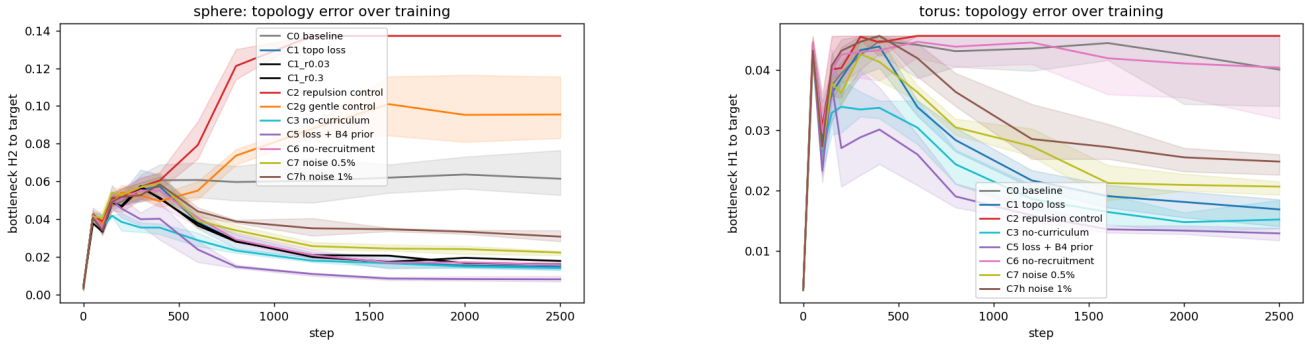


Figure 1. Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min-max). Left, sphere (H2): every loss arm (C1, C3, C5, C6, both ρ variants, noise pair C7/C7h) converges to a fraction of baseline; the norm-matched repulsion control C2 is destructive, and its gentle variant C2g still ends worse than baseline. Right, torus (H1), the class no prior won: same ordering, C5 best; the noise arms descend shallower but never approach baseline — except the no-recruitment ablation C6, which rides it.

loops still exist ($\#sig$ H1 = 2 in every seed; Fig. 1, right: the C6 curve rides the baseline). The run logs locate the mechanism: one torus loop is significant at every refresh while the other sits chronically below the significance threshold (all five seeds, all 200 refreshes), so matching and diagonal pressure act only on the loop the live diagram already measures — recruitment is the sole gradient path to the other; on this class it is the mechanism, not a cold-start device. On the sphere the void is always live and matched, recruitment never fires (0 of 200 refreshes, all seeds), so C6 is loss-identical to C1 by construction; its tail (.0166 vs .0156, 1.5 σ) doubles as a measurement of pure run-to-run (CUDA) noise.

Sensor-noise stress (C7/C7h). With the plan’s cloud perturbed at every refresh ($\sigma = 0.5\%/1\%$ of the diagonal $\approx 0.5/1.0$ median sample spacings, §4), the loss degrades gradually and never harms: torus .0210 $\pm$.0018 / .0259 $\pm$.0022 (2.0 $\times$ /1.6 $\times$ below baseline, vs C1’s 2.3 $\times$), sphere .0235 $\pm$.0009 / .0328 $\pm$.0013 (2.7 $\times$ /1.9 $\times$, vs 4.0 $\times$) — at equal-or-better Chamfer, with the correct significant-feature count in every run (Fig. 1). Internals: across all 2,400 noisy refreshes not one significant pair failed the Gabriel gate, and recruitment

absorbed the noise (at $\sigma=1\%$ the torus’s second loop is sub-threshold at every refresh and recruitment reaches it every time, zero unreached targets). Calibration is the noisiest part (λ_{peak} spreads $\approx 3\times$ across seeds), yet tails stay tight — consistent with the flat ρ response.

5.2. Generality: external genus-known meshes

Everything above is on one analytic family, so we repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ and ramp unchanged, density fixed by the floor rule (§4) — on four external genus-known meshes: spot (0, organic), bob (1, organic), fandisk (0, CAD), and the tom-yum pot (3, artist-authored, certified by ingest; §4). No per-shape tuning of any kind. All four pass the verdict rule at equal-or-better Chamfer (Table 2; trajectories in suppl. Fig. S2), and the control’s failure modes replicate exactly: repulsion erases the void on all three H2 shapes (β_2 1 $\rightarrow$ 0, all seeds; on the tom-yum pot at 1.36 $\times$ worse Chamfer) and collapses a loop on the genus-1 shape (β_1 2 $\rightarrow$ 1, all seeds).

Four observations. The H1 claim survives the family change: bob halves loop error with zero phantom handles in every seed (26.1 σ), replicating torus. Sharp-featured CAD benefits most: fandisk’s 10.4 $\times$ exceeds every analytic-family effect. Complex organic geome-

Table 2. Generality wave: tail bottleneck to target (mean $\pm$ sd, 3 seeds); $\times$ = reduction vs C0; σ = Welch. Same verdict rule as Table 1.

shape (dim)	C0	C1 loss	C2 control	verdict
spot (H2)	.0521 $\pm$.0081	.0237 $\pm$.0000 (2.2 $\times$)	.0652 (β_2 1 $\rightarrow$ 0)	PASS (6.1 σ)
bob (H1)	.0426 $\pm$.0012	.0208 $\pm$.0008 (2.1 $\times$)	.0437 (β_1 2 $\rightarrow$ 1)	PASS (26.1 σ)
fandisk (H2)	.0538 $\pm$.0022	.0052 $\pm$.0008 (10.4 $\times$)	.0571 (β_2 1 $\rightarrow$ 0)	PASS (35.7 σ)
tom-yum pot (H2)	.0221 $\pm$.0003	.0057 $\pm$.0002 (3.9 $\times$)	.0223 (β_2 1 $\rightarrow$ 0)	PASS (80.1 σ)



Figure 2. The tom-yum pot: an artist-authored Blender model (raw export: 9 open shells, 232 non-manifold junction edges) ingested and certified to a single genus-3 shell, $\beta=(1,6,1)$ exact (§4). Right: cutaway — the narrow flue mouth caps at small α , so the flue interior reads as an enclosed chamber: the on-axis H2 void the floor rule selects at $M=8192$ (all H1 loops sit below the floor).

try exposes a count-level limit: spot’s baseline manufactures a phantom void in two of three seeds (final #sig H2 = 2, 1, 2); the loss clears it in two seeds but retains one spurious bar in the third (1, 2, 1) — value repair is reliable, count repair on sub-budget organic detail (legs, horns at $N=1200$) is an improvement, not a guarantee; spot’s C1 tail is pinned across seeds (sd 0 at .0237) — a geometric floor of the double-torus kind, far milder. And the pipeline reaches raw artist content: the tom-yum pot (Fig. 2) extends the wave in two directions at once — input class (a raw Blender export, certified rather than constructed, §4) and observable (an H2 void clearing the measurement floor only at $M=8192$) — both chosen by the pre-registered rule, not by hand: the dress rehearsal for scan data, whose ground truth must likewise be certified. The loss cuts the void’s value error 3.9 $\times$ at Chamfer parity (0.99 $\times$ baseline); the 80.1 σ Welch statistic reflects an unusually tight seed spread at $n=3$ (C1 sd .0002), so the honest summary is non-overlapping seed ranges, with the effect size inside the study’s 2.1–10.4 $\times$ span. The control destroys the void (β_2 1 $\rightarrow$ 0 in all three seeds) at 1.36 $\times$ worse Chamfer; its bottleneck pins at the unmatched-target bound (the target bar’s half-life, .0223), so the count — not the meaningless value gap (0.8 σ) — is the verdict. Two caveats transfer: the base-

line already reads the correct count (a value-accuracy win, like bob), and ground truth is certified, not constructed.

6. Mechanism and discussion

Sparse tugs vs. budget allocation. Each significant pair back-propagates through the 3–7 sample positions of its birth and death simplices: precise but sparse pulls, iterated by re-pairing — the toys show surgery “walking” around a defect a few points at a time; the resampling prior is the opposite — broad, indirect, budget-limited. The results sort cleanly along this axis: 1-D loops, which allocation can only over- or under-tessellate, respond to value pressure (H1 win); wide 2-D shells benefit from both (prior helps, loss helps more, composition best); 0-D structure needs neither (H0 null, three channels).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface: the void’s death pins at that equilibrium (identical across seeds), then drags, while near-diagonal phantom bars accumulate (diagram trajectories: suppl. Fig. S1), and thin structures collapse (β_2 on the cube, β_1 on the torus). The loss has no length-scale constraint — it pulls the death coordinate itself and the geometry follows. This sharpens the control argument: spending the same gradient budget through the same channel without topological information is not neutral but harmful, so C1’s wins cannot be an artifact of extra regularization.

Measurability at the loss’s density. A persistence loss can only enforce features that clear the significance floor at its sampling density M : the torus void and the double torus’s two tube loops do not, at $M=2048$. We treat this as a principle, not a nuisance: the target bundle is built (and audited) at the same M , features below the floor are excluded from the objective, and the double torus is evaluated against its two measurable loops — also all the reconstruction itself can express at feasible budgets.

The floor is quantifiable: a bar is significant iff its lifetime exceeds $6r_{\text{med}}(M)$ (r_{med} = median nearest-neighbour spacing, the implementation’s self-calibrating threshold), and r_{med} follows the uniform-sampling law $\propto M^{-1/2}$ (measured .0090→.0028 over $M=2048 \rightarrow 20000$; ratio 3.16 vs $\sqrt{9.77}=3.13$). The double torus’s tube loops live at lifetime .045–.046 on the clean surface: at $M=2048$ that is $0.83\text{--}0.85\times$ the floor — invisible by a 15% margin, clearing it only from $M \approx 3 \times 10^3$ — while at the evaluation density ($M=20000$, floor .0171) they clear it $3.1\times$; the binding constraint there is the representation (the soup never expresses the tubes), so raising M alone cannot close the gap. The torus quantifies the interaction with noise: its second loop clears the clean floor by only $1.33\times$, a margin the live cloud’s noise consumes — why it is chronically sub-threshold in training and recruitment is its only gradient path (C6). Recruitment is thus how the loss reaches below its own significance floor on the live side — landscape-guided in practice (unproductive tears self-extinguish at the local detour scale; suppl. §C), with the location-blindness caveat standing for target features larger than any achievable detour. Suppl. Table S3 tabulates the counts across M (at 4096 all four double-torus loops clear). The tom-yum pot ran the rule prospectively (§4): it picked observable class and bundle density before any training run; the in-loop result (Table 2) validated it. Adaptive or multi-scale density is the natural extension; the trade is explicit, since persistence cost grows $O(M \log M)$.

Prescription. Correct topology in the loss; allocate with a wide prior; combine. Curriculum schedules are unnecessary once the loss scale is calibrated against the photometric gradient.

7. Limitations

Synthetic, closed, single-machine. All quantitative claims are on synthetic renders of closed surfaces with known topology, on one GPU: an analytic family plus four external genus-known meshes (§5.2; one a raw artist mesh ingested and certified — the intended path for scans), still short of a public benchmark. A real (open-surface, clinical) demonstration remains future work: scan boundaries contaminate H1 and no clean ground-truth diagram exists; of the concrete paths (boundary-aware persistence, filtering boundary-born bars, template-aligned initialization) we judge bar-filtering the pragmatic near-term step, with a synthetic open-surface stress test first.

Density floor. The loss cannot see features below the significance threshold at its sampling density (torus

void; two of four double-torus loops at $M=2048$); the bundle audit makes this visible, not invisible — but it is a real ceiling on what the objective can enforce.

Fixed target diagram. The target bundle presumes known ground-truth topology (true where a prior scan or template exists; not in blind capture). Template-free designs are natural but untested: minimizing total persistence (risks collapsing legitimate structure), a target-genus penalty on the significant-H1 count (what we would try first), or statistical priors over a shape class’s diagrams.

Control strength. The repulsion control is aggressive (it destroys topology at the calibrated magnitude); a gentler variant at half the radius is still $1.6\times$ worse than baseline (sphere: .0972 vs .0623), so the verdicts do not hinge on control tuning — but control design deserves more variety than one family at two strengths.

Nondeterminism. DiffSoup’s CUDA training is not bit-reproducible: divergence enters at resampling, where borderline prune/split decisions flip under floating-point drift. Tail coefficients of variation run 7–10% for baseline and loss arms alike, and a loss-identical re-run pair (the sphere C6 arm, which is C1 by construction there) landed 6% apart — the scale of pure run-to-run noise. All statements are seed-averaged with Welch statistics.

Recruitment guarantees. Recruitment’s self-correction premise — unproductive edits saturate at the local detour scale — is validated empirically (toys, training, stress test), not proven; pathological geometries (thin filaments, tightly constrained manifolds) could defeat it; a failure-case study and a hardness analysis are future work.

8. Conclusion

We moved topology out of the resampler’s allocation channel into the training objective. Four things carry it: a matched persistence loss whose pair-frozen backward is certified per pairing by a Gabriel-exactness gate; a recruitment term restoring the gradient optimal matching provably lacks for missing features — the load-bearing mechanism for loops; a channel-controlled evaluation where every gain must beat a norm-matched non-topological control at Chamfer parity; and a quantified account of the floors (measurement, $6r_{\text{med}}(M)$; representation, budget N). The verdicts replicate on external genus-known meshes — including a raw artist mesh certified by measurement — without per-shape tuning, and degrade gracefully under noise. The prescription: correct topology in the loss, allocate with a wide prior, combine. Future work: open surfaces and real scans, adaptive sampling density, template-free targets, curvature-weighted masking.

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